



Comunicações Móveis

1º Semestre, 2026/2027

Daniel Corujo
Francisco Fontes

Mobile Networks

- Professors

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	Segunda	Terça	Quarta	Quinta		Sexta
9:00			CM 04.3.30 dcorujo@ua.pt Daniel Nunes Corujo TP3 (TP)	CM 04.3.30 dcorujo@ua.pt Daniel Nunes Corujo TP1 (TP)	CM 04.3.31 fontes@ua.pt Francisco Manuel Marques Fontes TP2 (TP)	
9:30						
10:00						
10:30						
11:00						
11:30						
12:00						



Planning

- 14 weeks scheduled for theoretical and practical classes
 - Information in elearning...
 - English language in slides
- Each class
 - ~50% of time: Theoretical and preparation for the...
 - ~50% of time: Practical work
- Each practical work
 - A step-by-step guide (with some “thinking” points for students)

Objectives

- Develop concepts associated to different types of mobile networks
 - Overall perspective of different wireless networks
 - Integration into a future vision, with heterogeneous environments.
 - “5G”-alike world.
 - Focus on networking, system and protocol aspects
 - IT-integration aspects mostly in other subject
 - Students should be able to:
 - understand new technologies and concepts of wireless communications;
 - be able to use their knowledge to react to the current changes in wireless networks
 - understand future mobile network integrated systems;

Program

1. Mobile Communications: Multicast, Broadcast, Radio vs IP
2. Wireless network models: IEEE802 architecture, Wi-Fi, Bluetooth
3. (Wide) Wireless Sensor networks: ZigBee, LoRa, SigFox
4. Mobile Networks and associated systems, Satellites
5. Mobile Communications systems evolutions: the Future!

Evaluation criteria



- Theory: 50%
 - Exam at the normal season
 - Final exam (a.k.a. exame de recurso) will contain all subjects as well
- Practical: 50%
 - 35%: Written mini-tests about the practical works and its subjects
 - 29 Out
 - 10 Dez
 - 15%: Special assignment (TBD)
- Minimum grade: 7

(Tentative) Plan



1	17/set	INTRO
2	24/set	WLAN (T) + WLAN 1 (P)
3	01/out	WLAN 1 (P) + WLAN 2 (P)
4	08/out	BT (T) + BT (P)
5	15/out	BT (P)
6	22/out	Redes Sensores/LoRa/ZigBee
7	29/out	TESTE1+IMS+GTP/Overlay + Redes Celulares + 5G 1
8	05/nov	5G 2
9	12/nov	5G 3
10	19/nov	5G 4
11	26/nov	5G 5
12	03/dez	MEC+Virtualização
13	10/dez	TESTE2, Finalização de Projetos
14	17/dez	APRESENTAÇÃO TRABALHOS

Organization

- All information to be displayed in e-learning
 - Announcements
 - Classes handout
 - Practical works
 - Evaluation and grades
- Summaries in PACO.